

Alexandre Champagne-Ruel

NASA Postdoctoral Program Fellow

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Origins of Life • Astrobiology • Complex Systems

RESEARCH PROFILE

NASA Postdoctoral Fellow and physicist investigating the physical principles underlying the origin of life. My research integrates statistical mechanics, information theory, and evolutionary dynamics to understand how spatial environments drive the emergence of complexity. Currently developing quantitative frameworks for agnostic life detection and biosignature assessment.

ACADEMIC POSITIONS

NASA Postdoctoral Program Fellow
Arizona State University, Mathis Group

2025 – Current
Tempe, AZ, USA

EDUCATION

Ph.D. Physics

2025

Université de Montréal

Montréal, QC, Canada

- Thesis: *Spatiality in Prebiotic Evolution: Toward a Physics of the Emergence of Complexity*
- Advisor: Paul Charbonneau

M.Sc. Physics

2020

Université de Montréal

Montréal, QC, Canada

- Thesis: *From Game Theory to Exobiology: The Emergence of Cooperation as a Critical Phenomenon*
- Advisor: Paul Charbonneau

B.Sc. Physics

2018

Université de Montréal

Montréal, QC, Canada

B.Sc. Philosophy

2012

Université de Montréal

Montréal, QC, Canada

GRANTS & AWARDS

Only individual grants >\$5,000 are listed.

NASA Postdoctoral Program Fellowship NASA

146,496 USD

2025

Project: Mapping Molecular Complexity for Agnostic Life Detection

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2024

Globalink Research Award Mitacs Canada

6,000 CAD

2024

Google Cloud Research Grant Google

5,000 USD

2023 – 2026

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2023

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2022

Doctoral Scholarship Fonds de Recherche du Québec

70,000 CAD

2022

Excellence Award Université de Montréal

10,000 CAD

2018

Various Travel & Research Grants 15 awards

~30,000 CAD

2011 – 2025

PUBLICATIONS

Articles under Review

A. **Champagne-Ruel** and C. Mathis, “Spatial Patterning and Selection: How the Environment Shapes Molecular Complexity”, [10.48550/arXiv.2509.04547](https://arxiv.org/abs/10.48550/arXiv.2509.04547) (2025), (Submitted to PNAS Nexus).

Published Articles

S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origins of life: An integrated review. Part 2: Theoretical methods and emerging trends”, *Cell Reports Physical Science* **0**, [10.1016/j.xcrp.2026.103211](https://doi.org/10.1016/j.xcrp.2026.103211) (2026).

S. Rajamani, C. Mathis, S. Asche, D. Boulesteix, C. Blanco, O. Markovitch, A. **Champagne-Ruel**, et al., “Better origin(s) of life research through collaboration across disciplines”, *Cell Reports Physical Science* **0**, [10.1016/j.xcrp.2026.103220](https://doi.org/10.1016/j.xcrp.2026.103220) (2026).

A. **Champagne-Ruel**, S. Zakaib-Bernier, and P. Charbonneau, “Diffusion and pattern formation in spatial games”, *Physical Review E* **110**, 014301 (2024).

A. **Champagne-Ruel** and P. Charbonneau, “A Mutation Threshold for Cooperative Takeover”, *Life* **12**, 254 (2022).

S. Gelin, A. **Champagne-Ruel**, and N. Mousseau, “Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons”, *Nature Communications* **11**, 3977 (2020).

PRESENTATIONS

Invited Talks

Spatiality in Prebiotic Evolution

Queens College, City University of New York

2026

New York, USA

Physics of Complexity And Agnostic Life Detection

Trottier Institute for Research on Exoplanets

2025

Montréal, QC, Canada

From Emergent Complexity to Reliable Life Detection

Arizona State University

2024

Tempe, AZ, USA

Coopération, émergence et transitions

Université du Québec à Montréal

2022

Montréal, QC, Canada

Contributed Talks

From Emergent Complexity to Reliable Life Detection

Biennial European Astrobiology Conference (BEACON)

2025

Reykjavik, Iceland

Diffusion: An Overlooked Driver of Prebiotic Complexity

Astrobiology Science Conference (AbSciCon)

2024

Providence, RI, USA

Théorie de l'information et origine de la vie

90e Congrès de l'ACFAS

2023

Montréal, QC, Canada

A Mutation Threshold for Cooperative Takeover

Astrobiology Science Conference (AbSciCon)

2022

Atlanta, GA, USA

Cooperation: An Emergent Universal Feature at the Dawn of Life

Interdisciplinary Origin of Life Meeting for Early Career Researchers

2022

Montréal, QC, Canada

Mutation Favors the Emergence of Cooperation*Life and Space Poland***2021***Virtual/Szczecin, Poland***La criticité dans un système évolutif artificiel***Centre de recherche en astrophysique du Québec (CRAQ)***2019***Orford, QC, Canada***Posters****Mapping Molecular Complexity for Agnostic Life Detection***NASA Postdoctoral Program Virtual Symposium***2025***Virtual***L'émergence de la coopération via l'évolution de réseaux informationnels***90e Congrès de l'ACFAS***2023***Montréal, QC, Canada***Diffusion, structures spatiales et origine de la vie***90e Congrès de l'ACFAS***2023***Montréal, QC, Canada***OoLEN - The Origin of Life Early-career Network: Building the Community Needed to Solve the Problem***AbSciCon***2022***Atlanta, GA, USA***A Mutation Threshold for Cooperative Takeover***Gordon Research Seminar: Challenging Paradigms in Prebiotic Chemistry***2022***Oxnard, CA, USA***A Mutation Threshold for Cooperative Takeover***Gordon Research Conference: Environments for the Origins of Life and Habitability***2022***Oxnard, CA, USA***Les mutations favorisent la coopération en contexte évolutif***Centenaire, Département de Physique, Université de Montréal***2021***Montréal, QC, Canada***Mutation Favors the Emergence of Cooperative Behavior***Molecular Origins of Life Munich***2021***Virtual/Munich, Germany***Mutations Promote Cooperation in an Evolutionary Setting***XIXth ISSOL Conference***2021***Virtual***Self-Organized Criticality : A Prelude to Avalanche Models of Solar Flares***Space Climate 7 Symposium***2019***Orford, QC, Canada***SCHOOLS & SELECTIVE WORKSHOPS****Assembly Theory for Folded Matter***Santa Fe Institute***2025***Santa Fe, NM, USA***Information Driven States of Matter***University of Rochester***2024***Rochester, NY, USA***TEACHING EXPERIENCE****Mentor – Biocomputing Scholars program***Arizona State University***2025 – 2026***Tempe, AZ, USA*

- Guiding 3 undergraduate students in developing computational chemistry simulations.

Undergraduate Internship Supervision*Université de Montréal***2022***Montréal, QC, Canada*

- Supervised a summer research project on diffusion; resulted in journal publication and poster presentation.

Teaching Assistant – Introduction to Astrobiology

Université de Montréal

2021 – 2022

Montréal, QC, Canada

Tutoring – Undergraduate Level

Université de Montréal

2018 – 2022

Montréal, QC, Canada

SERVICE & OUTREACH

Scientific Leadership

Positions

Member of the Executive Board

Origin of Life Early-career Network (OoLEN)

2022 – Present

Origin of Life Digest ([link](#))

Editor

2021 – Present

Conference Organization

Interdisciplinary Origin of Life Meeting

Organizing Committee

2026

Tokyo, Japan

Frontiers in Astrobiology and Origins of Life Conference

Organizing Committee

2025

Reykjavik, Iceland

Origine de la vie : de l'astrophysique à la philosophie

Lead Organizer

2023

Montréal, QC, Canada

Interdisciplinary Origin of Life Meeting for Early Career Researchers

Lead Organizer

2022

Montréal, QC, Canada

Space Climate 7

Local Organizing Committee

2019

Orford, QC, Canada

Session Convener

AbSciCon

Assembly Theory Across Scales: From Molecules to Planetary Systems

2026

Madison, WI, USA

AbSciCon

Exploring Self-Assembly and Self-Organization in Prebiotic Complexity

2026

Madison, WI, USA

Media & Public Engagement

É. Beaudoin-Paul, “Entrevue avec Alexandre Champagne-Ruel, récipiendaire de la prestigieuse bourse en astrobiologie de la NASA”, [Quartier Libre](#) (2025).

A. Riopel, “Comment reconnaître la vie sur d’autres planètes”, [Le Devoir](#) (2023).

C. Tamura, “Cooperation and the Origin of Life”, Quantum Photonics Clubhouse Podcast (2022).

Reviewing Activities

- Journal of the Royal Society Open Science
- Journal of the Royal Society Interface
- Trends in Biochemical Sciences

Memberships

- International Society for Artificial Life
- Center for Research in Astrophysics of Québec
- Canadian Association of Physicists
- Canadian Astronomical Society
- Origin of Life Early-career Network
- International Society for the Study of the Origin of Life
- Complex Systems Society
- Scientific Society for Astrobiology (Founding Member)

TECHNICAL SKILLS

- **Languages:** Fluent in Spoken/Written French, English
- **Programming:** Python, C++, Fortran, Julia, R, LaTeX, MatLab, Assembly, Bash
- **Modeling:** Agent-based, Evolutionary Algorithms, Machine Learning, Network Theory, Game Theory
- **Development Workflow:** Linux, High Performance Computing (HPC)/SLURM, Git, Jekyll, Hugo, HTML5/CSS3